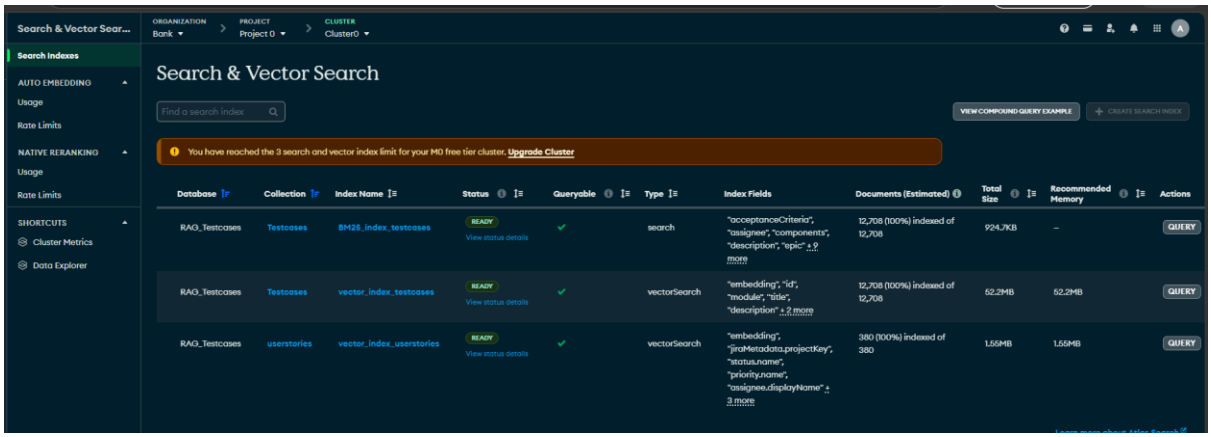
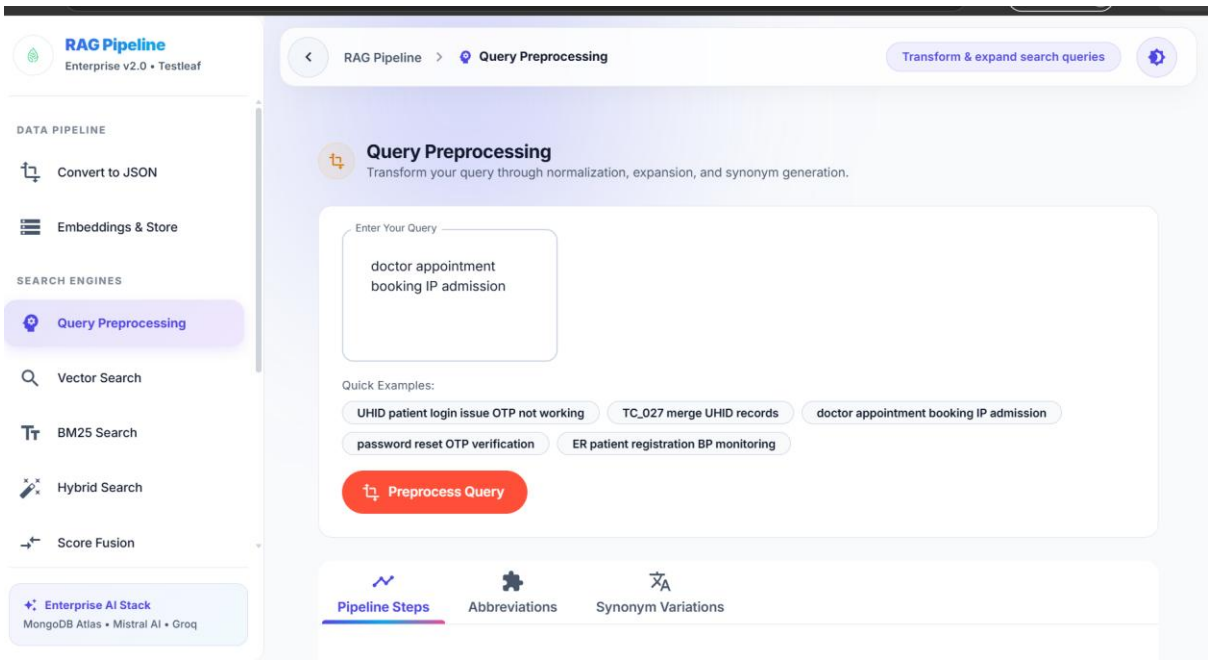
Home Assignment:

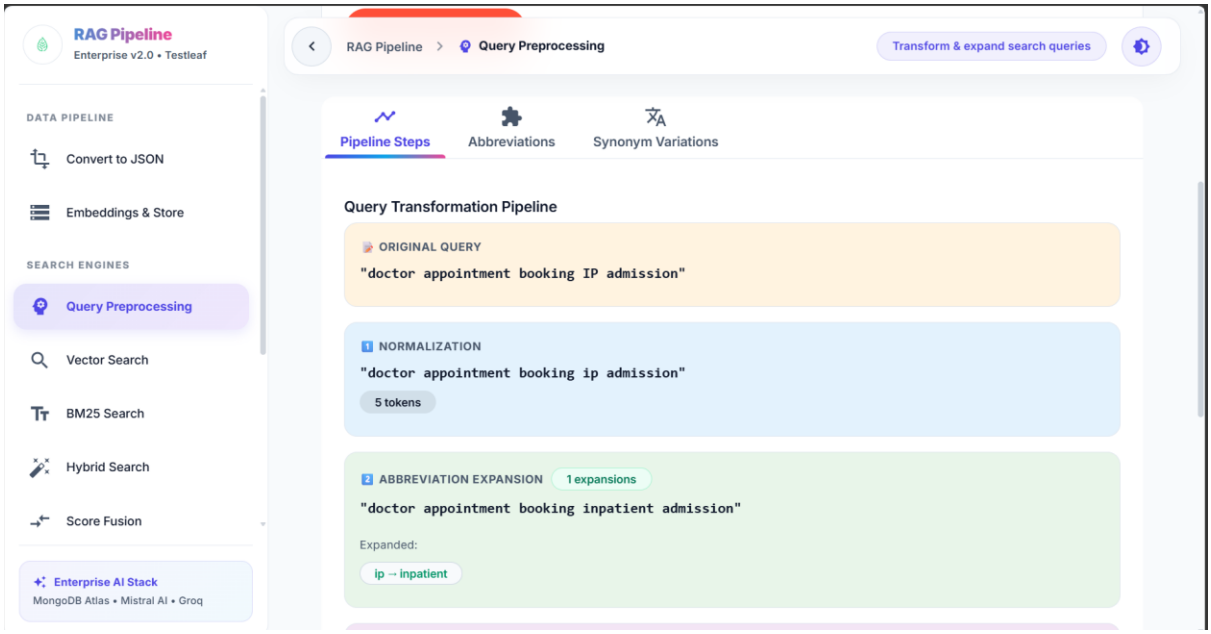
Create a BM25 index and explore till hybrid search.

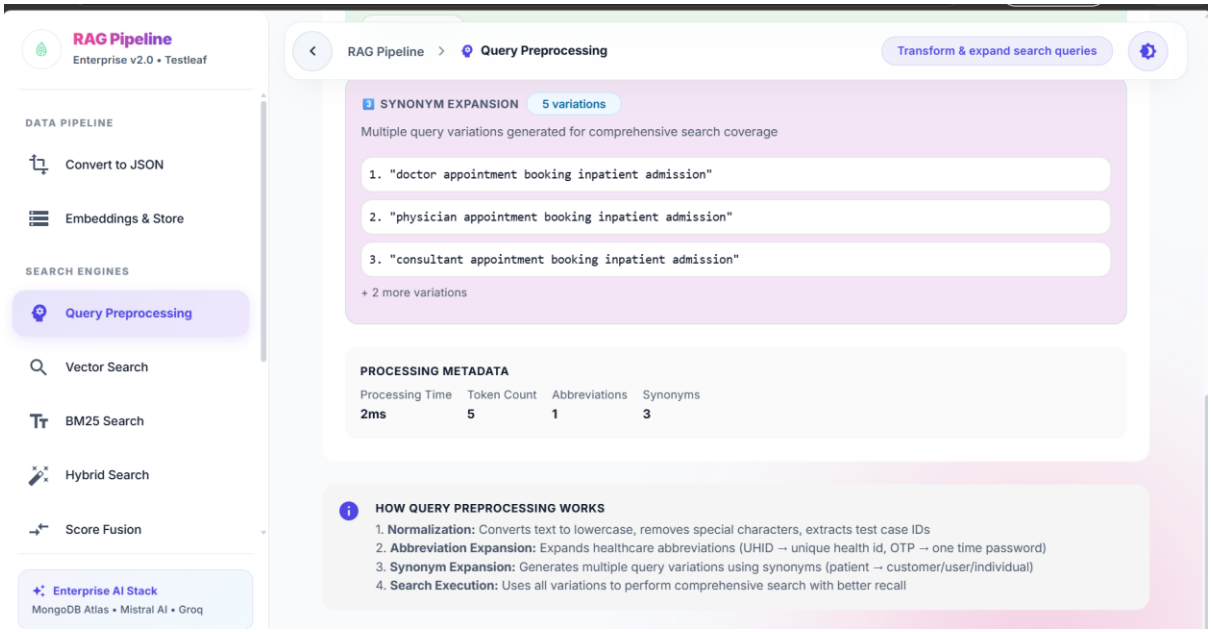
Created BM25 index in Mongo DB



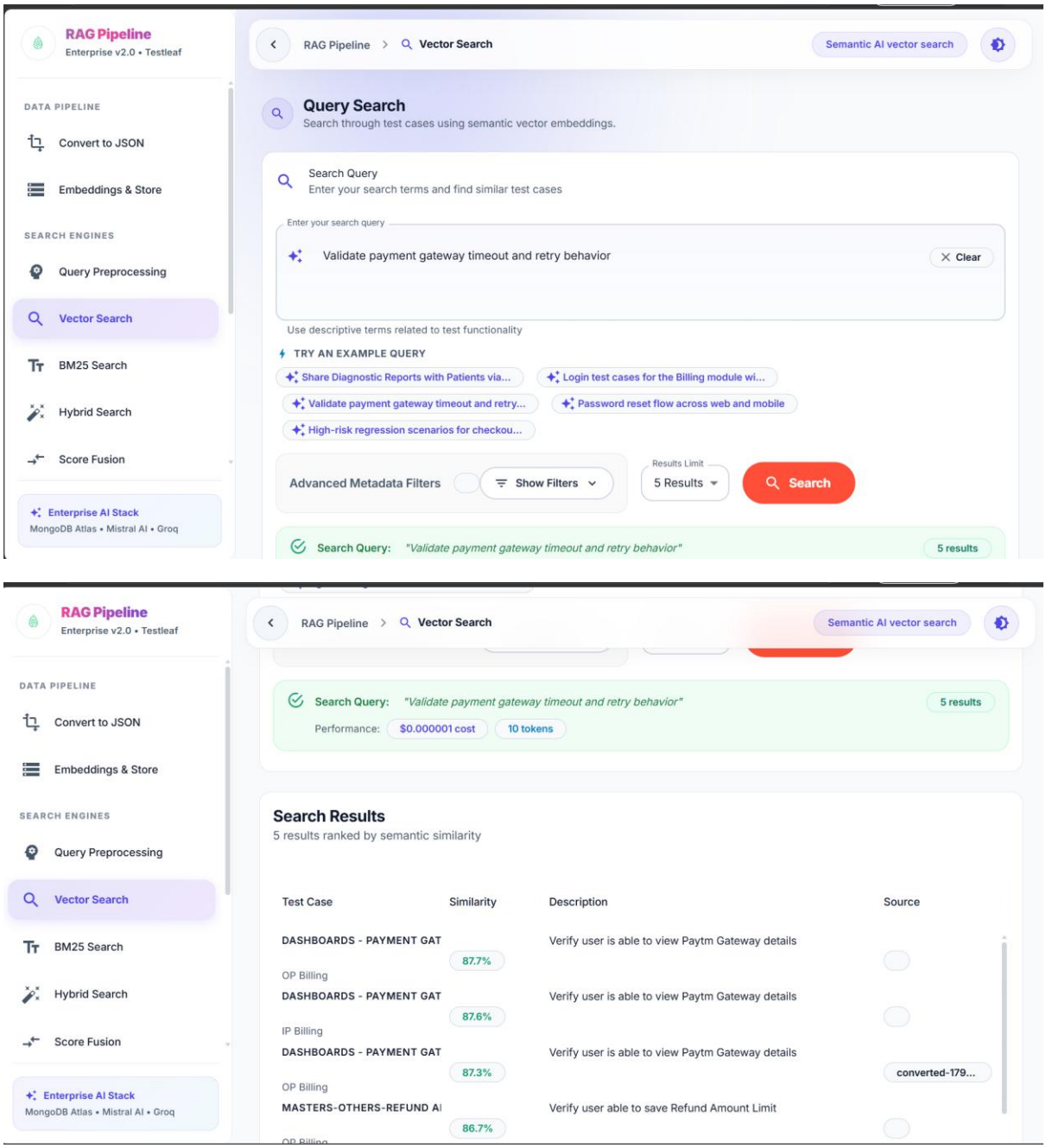
Query Processing

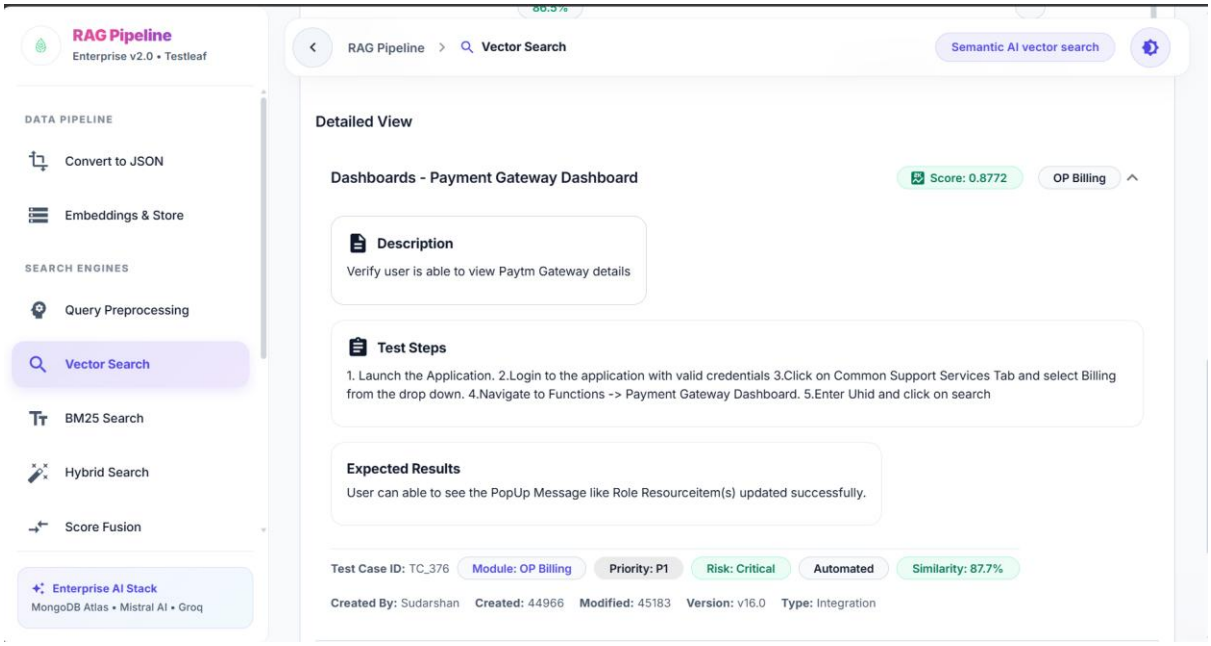


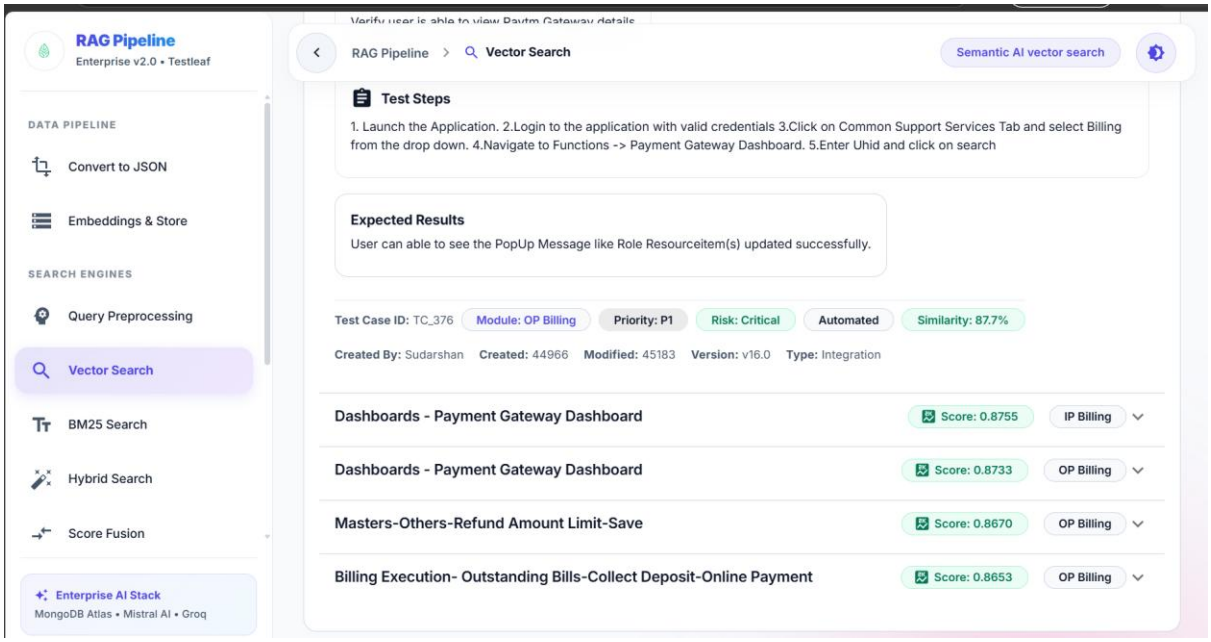

RAG Pipeline
Enterprise v2.0 • Testleaf
DATA PIPELINE
Convert to JSON
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Score Fusion
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Query Preprocessing
Transform & expand search queries
Pipeline Steps
Abbreviations
Synonym Variations
Query Transformation Pipeline
ORIGINAL QUERY
"doctor appointment booking IP admission"
NORMALIZATION
"doctor appointment booking ip admission"
5 tokens
ABBREVIATION EXPANSION
1 expansions
"doctor appointment booking inpatient admission"
Expanded:
ip → inpatient


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RAG Pipeline
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Transform & expand search queries
SYNONYM EXPANSION
5 variations
Multiple query variations generated for comprehensive search coverage
1. "doctor appointment booking inpatient admission"
2. "physician appointment booking inpatient admission"
3. "consultant appointment booking inpatient admission"
+ 2 more variations
PROCESSING METADATA
Processing Time
2ms
Token Count
5
Abbreviations
1
Synonyms
3
HOW QUERY PREPROCESSING WORKS
1. Normalization: Converts text to lowercase, removes special characters, extracts test case IDs
2. Abbreviation Expansion: Expands healthcare abbreviations (UHID → unique health id, OTP → one time password)
3. Synonym Expansion: Generates multiple query variations using synonyms (patient → customer/user/individual)
4. Search Execution: Uses all variations to perform comprehensive search with better recall

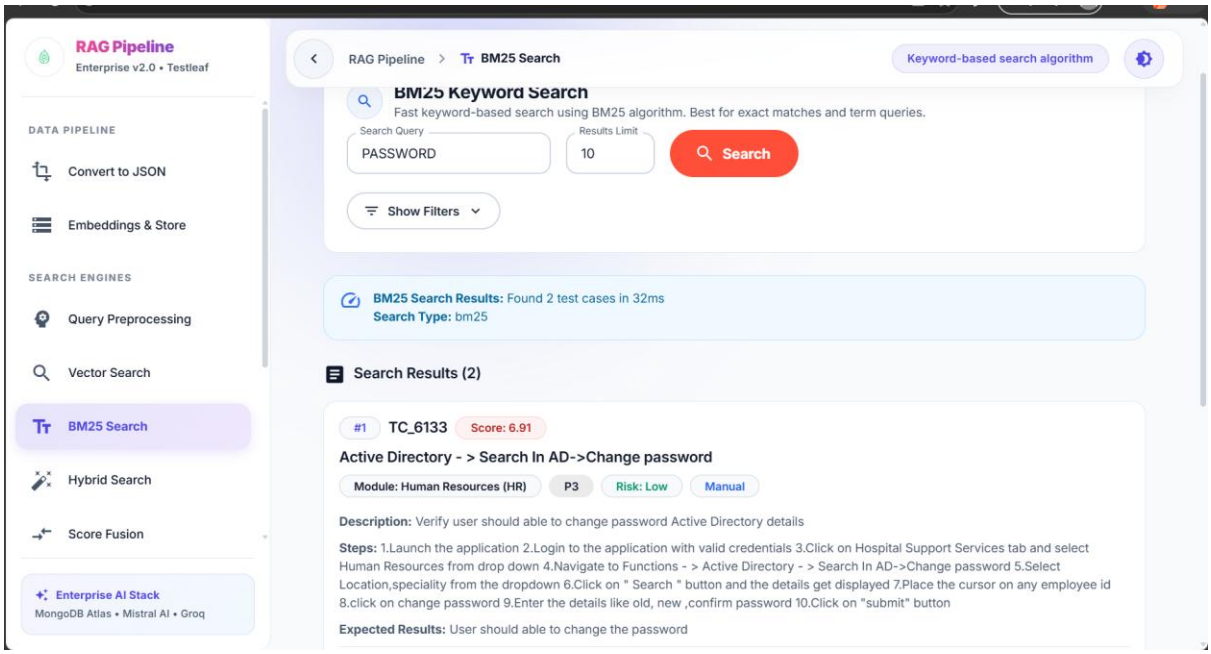
Vector Search

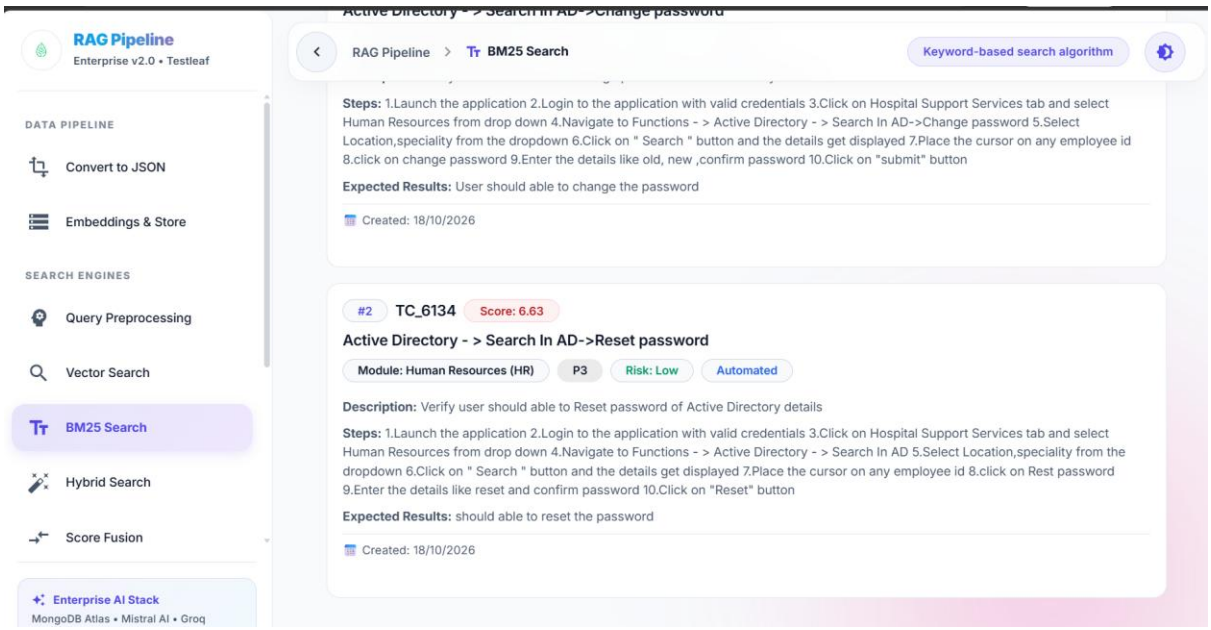



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Vector Search
Semantic AI vector search
Detailed View
Dashboards - Payment Gateway Dashboard
Score: 0.8772
OP Billing
Description
Verify user is able to view Paytm Gateway details
Test Steps
1. Launch the Application. 2.Login to the application with valid credentials 3.Click on Common Support Services Tab and select Billing from the drop down. 4.Navigate to Functions -> Payment Gateway Dashboard. 5.Enter Uhid and click on search
Expected Results
User can able to see the PopUp Message like Role Resourceitem(s) updated successfully.
Test Case ID: TC_376
Module: OP Billing
Priority: P1
Risk: Critical
Automated
Similarity: 87.7%
Created By: Sudarshan
Created: 44966
Modified: 45183
Version: v16.0
Type: Integration

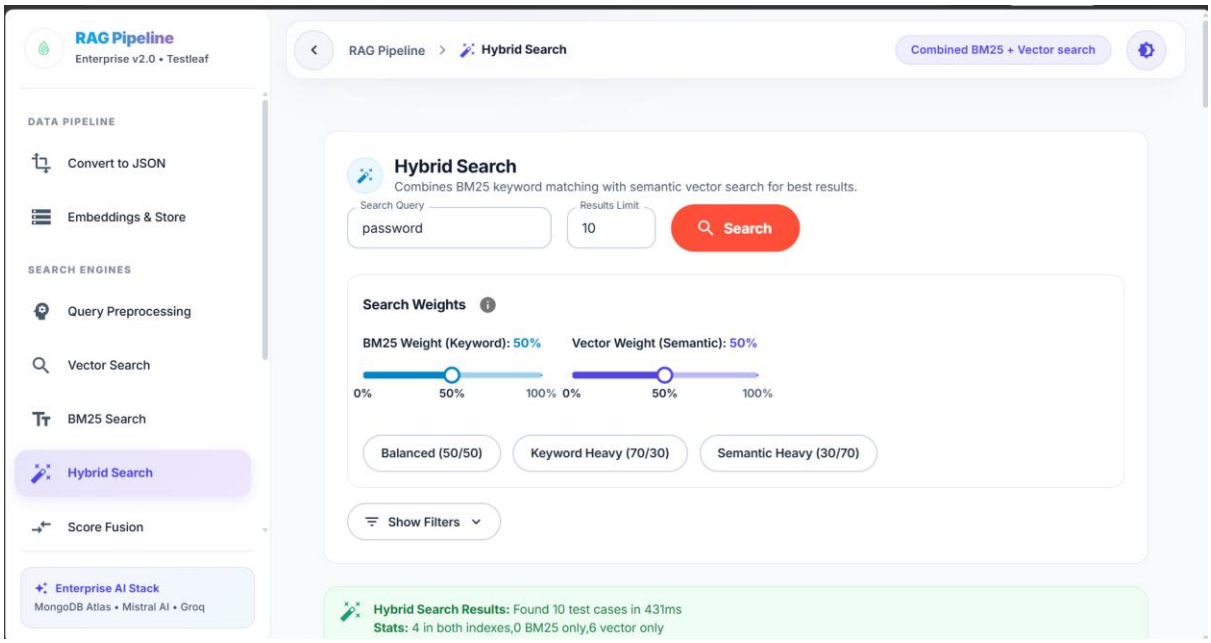

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RAG Pipeline
Vector Search
Semantic AI vector search
Test Steps
1. Launch the Application. 2.Login to the application with valid credentials 3.Click on Common Support Services Tab and select Billing from the drop down. 4.Navigate to Functions -> Payment Gateway Dashboard. 5.Enter Uhid and click on search
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Module: OP Billing
Priority: P1
Risk: Critical
Automated
Similarity: 87.7%
Created By: Sudarshan
Created: 44966
Modified: 45183
Version: v16.0
Type: Integration
Dashboards - Payment Gateway Dashboard
Score: 0.8755
IP Billing
Dashboards - Payment Gateway Dashboard
Score: 0.8733
OP Billing
Masters-Others-Refund Amount Limit-Save
Score: 0.8670
OP Billing
Billing Execution- Outstanding Bills-Collect Deposit-Online Payment
Score: 0.8653
OP Billing

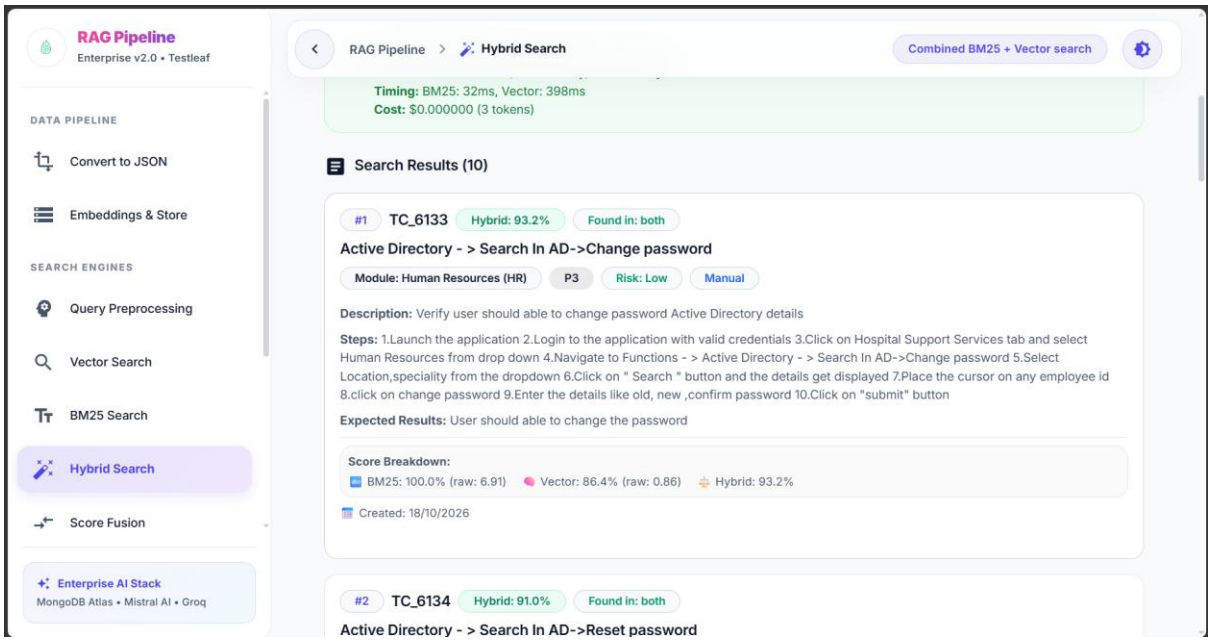
BM25 Search

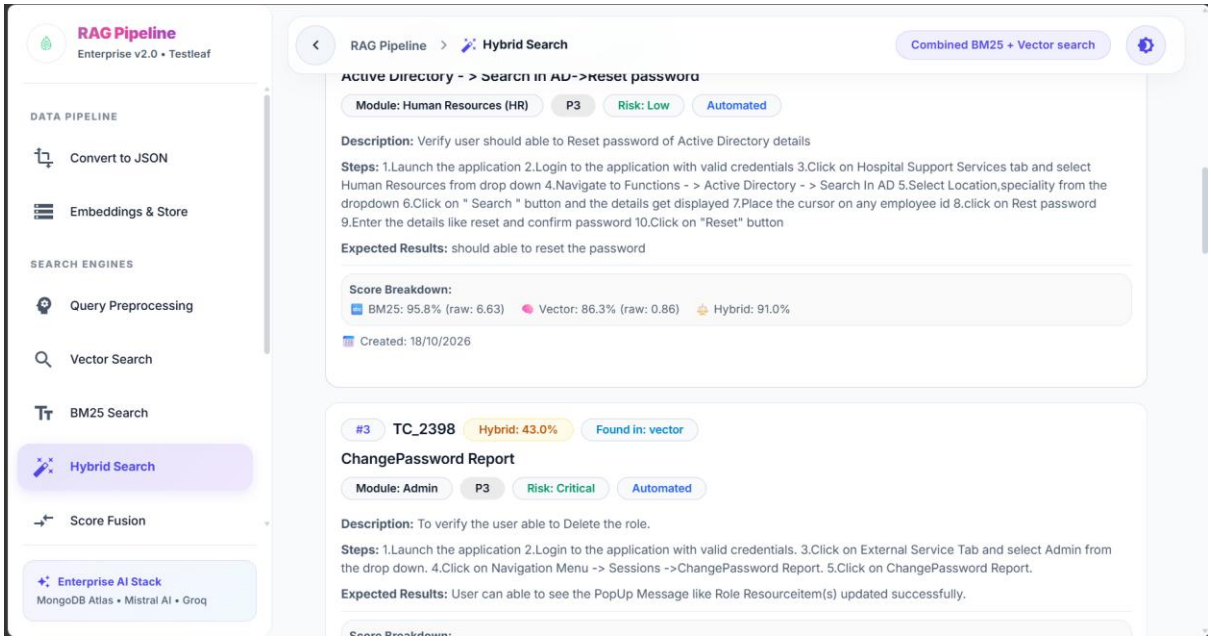

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RAG Pipeline > BM25 Search
Keyword-based search algorithm
BM25 Keyword Search
Fast keyword-based search using BM25 algorithm. Best for exact matches and term queries.
Search Query
PASSWORD
Results Limit
10
Search
Show Filters
BM25 Search Results: Found 2 test cases in 32ms
Search Type: bm25
Search Results (2)
#1
TC_6133
Score: 6.91
Active Directory - > Search In AD->Change password
Module: Human Resources (HR)
P3
Risk: Low
Manual
Description: Verify user should able to change password Active Directory details
Steps: 1.Launch the application 2.Login to the application with valid credentials 3.Click on Hospital Support Services tab and select Human Resources from drop down 4.Navigate to Functions - > Active Directory - > Search In AD->Change password 5.Select Location,speciality from the dropdown 6.Click on " Search " button and the details get displayed 7.Place the cursor on any employee id 8.click on change password 9.Enter the details like old, new ,confirm password 10.Click on "submit" button
Expected Results: User should able to change the password


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RAG Pipeline > BM25 Search
Keyword-based search algorithm
Steps: 1.Launch the application 2.Login to the application with valid credentials 3.Click on Hospital Support Services tab and select Human Resources from drop down 4.Navigate to Functions - > Active Directory - > Search In AD->Change password 5.Select Location,speciality from the dropdown 6.Click on " Search " button and the details get displayed 7.Place the cursor on any employee id 8.click on change password 9.Enter the details like old, new ,confirm password 10.Click on "submit" button
Expected Results: User should able to change the password
Created: 18/10/2026
#2
TC_6134
Score: 6.63
Active Directory - > Search In AD->Reset password
Module: Human Resources (HR)
P3
Risk: Low
Automated
Description: Verify user should able to Reset password of Active Directory details
Steps: 1.Launch the application 2.Login to the application with valid credentials 3.Click on Hospital Support Services tab and select Human Resources from drop down 4.Navigate to Functions - > Active Directory - > Search In AD 5.Select Location,speciality from the dropdown 6.Click on " Search " button and the details get displayed 7.Place the cursor on any employee id 8.click on Rest password 9.Enter the details like reset and confirm password 10.Click on "Reset" button
Expected Results: should able to reset the password
Created: 18/10/2026

Hybrid Search






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Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Hybrid Search
Combined BM25 + Vector search
Active Directory - > Search in AD->Reset password
Module: Human Resources (HR)
P3
Risk: Low
Automated
Description: Verify user should able to Reset password of Active Directory details
Steps: 1.Launch the application 2.Login to the application with valid credentials 3.Click on Hospital Support Services tab and select Human Resources from drop down 4.Navigate to Functions - > Active Directory - > Search In AD 5.Select Location,speciality from the dropdown 6.Click on " Search " button and the details get displayed 7.Place the cursor on any employee id 8.click on Rest password 9.Enter the details like reset and confirm password 10.Click on "Reset" button
Expected Results: should able to reset the password
Score Breakdown:
BM25: 95.8% (raw: 6.63)
Vector: 86.3% (raw: 0.86)
Hybrid: 91.0%
Created: 18/10/2026
#3
TC_2398
Hybrid: 43.0%
Found in: vector
ChangePassword Report
Module: Admin
P3
Risk: Critical
Automated
Description: To verify the user able to Delete the role.
Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on External Service Tab and select Admin from the drop down. 4.Click on Navigation Menu -> Sessions ->ChangePassword Report. 5.Click on ChangePassword Report.
Expected Results: User can able to see the PopUp Message like Role Resourceitem(s) updated successfully.